

Analytical Attempt at the Coupled Diffusive–Advective Disk Problem: Where the Separation Argument Breaks, and the Correct Route Forward

Kevin Bedoya (analysis notes)

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Abstract

We revisit the eigenvalue formulation of the two-layer (diffusive layer DL + advective microtubule layer AL) problem on the wedge. We audit the existing derivation, accept every step through the elimination of the AL density as a functional $R[F]$ and the in-wedge Helmholtz separation, and identify the precise point at which the argument becomes inconsistent: the angular boundary conditions are miscounted. The microtubule (MT) condition “BC3” is not an extra constraint layered on top of periodicity; it *is* the second angular boundary condition (a derivative *jump*, not derivative continuity). Re-imposing the boundary data correctly, we derive the exact eigen-conditions for a separable ansatz and show that two structural obstructions forbid a single-product solution: an explicit $2/r$ geometric coefficient in BC3, and the fact that $R(r)$ is not proportional to the radial profile $\rho(r)$. This is the analytic counterpart of the numerical finding that the density is $g(r) + a(r) \cos N\theta$ rather than one product. We then set up the construction that does work — the Laplace/resolvent method — carry the per-mode radial Green’s function in closed form, and exhibit the transcendental secular determinant whose roots are the (generally complex) decay rates. A two-mode reduction consistent with the numerics gives an explicit leading-order dispersion relation.

1 Problem and notation

In the eigenmode ansatz $(\phi, \rho) = (F(r, \theta), R(r))e^{-\sigma t}$ the reduced wedge problem (problem statement, Sec. 2–3) is

$$-\sigma F = \nabla^2 F = F_{rr} + \frac{1}{r} F_r + \frac{1}{r^2} F_{\theta\theta}, \quad -\sigma R = v R' + a F(r, 0) - b R, \quad (1)$$

on $r \in [0, 1]$, $\theta \in [0, 2\pi/N]$, with

$$\begin{aligned} \text{(i)} \quad & F(1, \theta) = 0, & \text{(ii)} \quad & F(r, 0) = F(r, \frac{2\pi}{N}), \\ \text{(iii)} \quad & a F(r, 0) = \frac{2}{r} \partial_\theta F|_{\theta=0} + b R(r), & \text{(iv)} \quad & R(1) = 0, \end{aligned} \quad (2)$$

together with regularity at $r = 0$. Condition (iii) is the MT condition obtained by integrating (1) across the filament at $\theta = 0$, after using the $\theta \rightarrow -\theta$ symmetry $-\partial_\theta F|_{2\pi/N} = \partial_\theta F|_0$.

2 Audit of the existing derivation

Valid: elimination of the AL density. The R -equation in (1) is first order in r with the single condition $R(1) = 0$. With $F(r, 0)$ regarded as a source it integrates exactly,

$$R(r) = -\frac{a}{v} e^{\frac{b-\sigma}{v} r} \int_1^r e^{-\frac{b-\sigma}{v} r'} F(r', 0) dr' =: R[F](r). \quad (3)$$

This step is correct and we keep it. Substituting (3) into (iii) closes the problem on F alone.

Valid: in-wedge separation. In the open wedge $\theta \in (0, 2\pi/N)$ there is no source, so a separated trial $F = \rho(r)\Theta(\theta)$ in (1) gives, with separation constant $\mu = \nu^2$,

$$\Theta'' + \nu^2\Theta = 0, \quad \rho'' + \frac{1}{r}\rho' + \left(\sigma - \frac{\nu^2}{r^2}\right)\rho = 0. \quad (4)$$

The radial equation is Bessel's; the solution regular at the origin is

$$\rho(r) = J_\nu(\sqrt{\sigma}r), \quad (5)$$

and the absorbing condition (i) requires

$$J_\nu(\sqrt{\sigma}) = 0. \quad (6)$$

All of this is correct.

The flawed step. The derivation next writes “ $\Theta(0) = \Theta(2\pi/N) \Rightarrow \nu = nN$ ” and proceeds with the single Fourier mode $e^{inN\theta}$. A second-order angular ODE (4) requires *two* boundary conditions. The standard Fourier modes $e^{inN\theta}$ with $\nu = nN$ are selected by *smooth* periodicity — continuity of *both* the value *and* the first derivative across $\theta = 0 \equiv 2\pi/N$. But the MT does **not** impose derivative continuity: integrating across the filament produced a derivative *jump*, which is exactly condition (iii). Thus smooth periodicity already exhausts the two angular boundary conditions, and (iii) is then mistakenly treated as a third, extra condition. This double-count is the origin of:

- the over-determined relation eq. (thirdBC.long) in the problem statement, an r -dependent identity required to hold for all r with no free parameter left;
- the puzzle recorded in blue, “the σ 's are determined by the diffusive BC alone.” They appear so only because periodicity was used to fix $\nu = nN$ and (6) was then read as $\sqrt{\sigma} = \kappa_{nN,j}$, spending all the freedom before the physics of the MTs (condition (iii)) had been imposed.

We pick up precisely here and impose the correct pair of angular conditions: periodic *value* (ii) and the MT jump (iii).

3 Exact eigen-conditions for the separable ansatz

Write the angular factor in real form,

$$\Theta(\theta) = \alpha \cos(\nu\theta) + \beta \sin(\nu\theta), \quad \Theta(0) = \alpha, \quad \Theta'(0) = \nu\beta. \quad (7)$$

Periodic value (ii) gives

$$\alpha(1 - \cos \frac{2\pi\nu}{N}) = \beta \sin \frac{2\pi\nu}{N}. \quad (8)$$

For the MT condition (iii), note $R[F]$ in (3) is linear in $F(r, 0) = \alpha\rho(r)$, so $R(r) = \alpha\tilde{R}(r)$ with

$$\tilde{R}(r) := -\frac{a}{v} e^{\frac{b-\sigma}{v}r} \int_1^r e^{-\frac{b-\sigma}{v}r'} \rho(r') dr', \quad v\tilde{R}' = (b-\sigma)\tilde{R} - a\rho, \quad \tilde{R}(1) = 0. \quad (9)$$

Substituting $F = \rho\Theta$ into (iii) and dividing by the common $\rho(r)$,

$$\boxed{\alpha \left[a - \frac{b\tilde{R}(r)}{\rho(r)} \right] = \frac{2\nu\beta}{r}} \quad \text{for all } r \in [0, 1]. \quad (10)$$

Equation (10) is the heart of the matter, and it exposes two independent obstructions to a separable solution.

3.1 Obstruction 1: the geometric $1/r$

Even with *no* feedback ($b = 0$), (10) reads $\alpha a = 2\nu\beta/r$, which cannot hold for all r unless $\alpha a = 0$ and $\nu\beta = 0$. The explicit $2/r$ in BC3 — inherited from the $1/r^2$ angular term of the polar Laplacian and the $1/r$ in the $\delta(\theta - \theta_i)/r$ source — makes the effective angular “Robin coefficient” $\Theta'(0)/\Theta(0) = \frac{r}{2}(a - b\tilde{R}/(\alpha\rho))$ depend on r . A constant separation constant ν cannot meet an r -dependent angular condition.

3.2 Obstruction 2: R is not proportional to ρ

Restore $b \neq 0$. For (10) to hold one would need $a\rho(r) - b\tilde{R}(r) \propto \rho(r)/r$, i.e. $\tilde{R}$ expressible through ρ and ρ/r . But $\tilde{R}$ solves the first-order ODE (9); eliminating it, a separable solution would force $v\rho' = -\sigma\rho$, i.e. $\rho(r) \propto e^{-\sigma r/v}$ — incompatible with the Bessel form (5) except trivially. Concretely, the purely even branch ($\beta = 0$, which by (8) forces $\nu = nN$, including the angularly symmetric $n = 0$) collapses (10) to $a\rho = b\tilde{R}$, and the same elimination gives $v\rho' = -\sigma\rho$, hence $\rho \equiv 0$. **There is no nontrivial angularly symmetric separable eigenmode**; the advective exponential and the diffusive Bessel profile are simply different functions.

3.3 No single-Helmholtz-mode eigenfunctions

One might hope to rescue separability with a finite sum of Fourier modes at a common σ . This fails for an independent reason. Any genuine eigenmode obeys $-\sigma F = \nabla^2 F$, regularity, and $F(1, \theta) = 0$ for *all* θ ; expanding $F = \sum_n A_n J_{nN}(\sqrt{\sigma}r) e^{inN\theta}$ and using orthogonality of the $e^{inN\theta}$ forces

$$A_n J_{nN}(\sqrt{\sigma}) = 0 \quad \text{for every } n. \quad (11)$$

A nontrivial multi-mode solution would need $\sqrt{\sigma}$ to be a common zero of J_{nN} for two distinct orders. Bessel functions of the first kind of distinct (integer) order share *no* positive zeros (Bourget's hypothesis, proven), and for real order all zeros are real, so complex σ does not help. Hence at any admissible σ only one angular order survives — and §3.1–§3.2 already showed that the single surviving mode cannot satisfy the MT condition.

Conclusion of §3. The coupled operator admits *no eigenfunction whose spatial part is a single separated Helmholtz-Bessel mode*. This is the exact analytic statement behind the numerical result: the computed density is well described by $g(r) + a(r) \cos N\theta$ (a dominant angularly symmetric piece plus a small r -dependent N -fold modulation), i.e. a sum of structures, never one product. Separation of variables is not merely inconvenient here — it is inconsistent with the microtubule boundary condition.

4 The correct route: time-Laplace / resolvent construction

Because exponential-separable modes do not exist, we solve the *time-dependent* problem directly and read the decay rates off as poles. Let $\hat{\phi}(r, \theta, s) = \int_0^\infty \phi e^{-st} dt$ and likewise $\hat{\rho}$. With initial data $\phi(r, \theta, 0) = \phi_0(r)$ (angularly symmetric) and $\rho(\cdot, 0) = 0$, (1)–(2) transform to

$$(\nabla^2 - s)\hat{\phi} = -\phi_0(r), \quad \hat{\rho}(r, s) = -\frac{a}{v} e^{\frac{b+s}{v}r} \int_1^r e^{-\frac{b+s}{v}r'} \hat{\phi}(r', 0, s) dr', \quad (12)$$

with $\hat{\phi}(1, \theta, s) = 0$, periodic value, and the transformed MT condition

$$a \hat{\phi}(r, 0, s) = \frac{2}{r} \partial_\theta \hat{\phi}|_0 + b \hat{\rho}(r, s). \quad (13)$$

(Here $\sigma \rightarrow -s$ relative to §3, so the radial functions below use the *modified* Bessel argument when $s > 0$.)

4.1 Per-mode radial Green's function

Expand in the angular Fourier basis allowed by periodic value, $\hat{\phi} = \sum_{n \in \mathbb{Z}} \hat{\phi}_n(r, s) e^{inN\theta}$. Each coefficient solves the inhomogeneous radial equation

$$\hat{\phi}_n'' + \frac{1}{r} \hat{\phi}_n' - \left(s + \frac{(nN)^2}{r^2}\right) \hat{\phi}_n = -\phi_0(r) \delta_{n0}, \quad \hat{\phi}_n(1, s) = 0, \quad \text{regular at } 0. \quad (14)$$

The homogeneous solutions are the modified Bessel functions $I_{nN}(\sqrt{s}r)$ (regular at 0) and $K_{nN}(\sqrt{s}r)$ (regular at ∞). The Green's function satisfying regularity at $r = 0$ and $G_n(1, r'; s) = 0$ at $r = 1$ is the

standard Sturm–Liouville construction

$$G_n(r, r'; s) = \frac{1}{W(r')} \frac{u_n^0(r_<) u_n^1(r_>)}{u_n^0(r) u_n^1(r)}, \quad \begin{aligned} u_n^0(r) &= I_{nN}(\sqrt{s} r), \\ u_n^1(r) &= I_{nN}(\sqrt{s} r) K_{nN}(\sqrt{s}) - K_{nN}(\sqrt{s} r) I_{nN}(\sqrt{s}), \end{aligned} \quad (15)$$

where u_n^1 is the homogeneous solution vanishing at $r = 1$, $r_< = \min(r, r')$, $r_> = \max(r, r')$, and the Wronskian combination $W(r') = r' [u_n^0 u_n^{1'} - u_n^{0'} u_n^1](r')$ is constant in r' (equal to u_n^0 : a θ -independent constant by Abel's identity). For the only forced mode ($n = 0$),

$$\hat{\phi}_0(r, s) = \int_0^1 G_0(r, r'; s) \phi_0(r') dr', \quad (16)$$

which is exact and closed in modified Bessel functions.

4.2 The microtubule condition couples the modes; the secular determinant

Equation (13) is imposed on the ray $\theta = 0$, where $\hat{\phi}(r, 0, s) = \sum_n \hat{\phi}_n(r, s)$ and $\partial_\theta \hat{\phi}|_0 = \sum_n inN \hat{\phi}_n(r, s)$. Substituting (12) for $\hat{\phi}$ (a linear functional of $\hat{\phi}(\cdot, 0, s)$) turns (13) into a single linear integro-functional relation among the radial coefficients $\{\hat{\phi}_n\}$, holding for all r . Projecting it onto the radial eigenbasis $\{J_{nN}(\kappa_{nN,j} r)\}_j$ (the Dirichlet basis of (14)) yields an infinite linear system

$$\mathbf{M}(s) \mathbf{c} = \mathbf{f}(s), \quad (17)$$

where $\mathbf{c}$ collects the modal/radial amplitudes, $\mathbf{f}$ is the $n = 0$ forcing from (16), and the matrix $\mathbf{M}(s)$ encodes the a , b/v , and $2inN/r$ couplings together with the AL integral kernel $e^{(b+s)(r-r')/v}$. The solution $\mathbf{c}(s) = \mathbf{M}(s)^{-1} \mathbf{f}(s)$ is meromorphic in s ; the physical density is recovered by Bromwich inversion

$$\phi(r, \theta, t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} \hat{\phi}(r, \theta, s) ds = \sum_k \text{Res}_{s=s_k} [e^{st} \hat{\phi}], \quad (18)$$

the sum running over the roots of the *secular determinant*

$$\boxed{\det \mathbf{M}(s) = 0} \quad (19)$$

together with any branch contribution. The roots $s_k = -\sigma_k$ are the decay rates; they are generally complex (the operator is non-self-adjoint because of the advection v and the one-sided MT coupling), and they are *not* the Bessel zeros $\kappa_{nN,j}^2$ — those reappear only as the poles of the *uncoupled* ($a = b = 0$) Green's function. This is the rigorous replacement for the (non-existent) separable spectrum.

5 A tractable leading-order reduction (guided by the numerics)

The numerics show that two angular modes ($n = 0$ and $n = 1$, order N) carry 0.9998 and 0.0002 of the energy, with all higher modes negligible. Truncating (17) to $n \in \{0, \pm 1\}$ and to the lowest radial Dirichlet function in each order gives a small explicit system. Writing $\hat{\phi}_0 \approx c_0 J_0(\kappa_{0,1} r)$ and $\hat{\phi}_{\pm 1} \approx c_{\pm 1} J_N(\kappa_{N,1} r)$ and projecting (13) onto these two radial profiles produces a 2×2 secular block

$$\det \begin{bmatrix} a p_{00}(s) - b q_{00}(s) & a p_{0N}(s) - b q_{0N}(s) \\ a p_{N0}(s) - b q_{N0}(s) - \frac{2N}{\cdot} \ell_N(s) & a p_{NN}(s) - b q_{NN}(s) \end{bmatrix} = 0, \quad (20)$$

where $p_{mn}(s) = \int_0^1 J_{mN}(\kappa_{mN,1} r) J_{nN}(\kappa_{nN,1} r) dr$ are overlap integrals, $q_{mn}(s)$ are the same overlaps weighted by the AL kernel from (12), and $\ell_N(s)$ carries the $2inN/r$ angular-jump term. The dominant (slowest) root $s_*(a, b, v, N)$ of (20) is the leading decay rate, and its eigenvector fixes the ratio $c_{\pm 1}/c_0 = a(r)$ amplitude observed numerically (small, growing toward the rim because $J_N(\kappa_{N,1} r)$ is pushed outward by its centrifugal N^2/r^2 barrier). This reproduces, analytically and to leading order, the $g(r) + a(r) \cos N\theta$ structure measured by the affine-overlap test, and reduces the eigenvalue to a single transcendental equation that can be solved numerically for s_* .

6 Summary

- Steps accepted as valid: the AL elimination $R[F]$ (3) and the in-wedge Helmholtz separation (5)–(6).
- Pick-up point: immediately after (6). The step $\nu = nN$ from periodicity is where the argument fails, because it spends both angular boundary conditions on *smooth* periodicity, leaving the MT condition (iii) as a phantom extra constraint.
- Imposing periodic value (ii) and the MT jump (iii) as the two angular conditions yields the master relation (10), which cannot hold for any single separated mode: the $2/r$ geometry makes the angular eigenvalue r -dependent, and $R \not\propto \rho$. With Bourget’s theorem ruling out multi-mode single- σ states, the coupled operator has no separable exponential eigenfunctions.
- The exact solution is therefore non-separable and is delivered by the Laplace/resolvent construction: a closed modified-Bessel Green’s function per angular mode (15)–(16), coupled through the MT condition into the secular determinant (19), whose roots are the (complex) decay rates.
- A two-mode reduction (20), justified by the measured energy budget, gives an explicit leading-order dispersion relation and recovers the numerically observed $g(r) + a(r) \cos N\theta$ form.

Honest status. A closed-form $\phi(r, \theta, t)$ is not attainable in elementary functions; the obstruction is structural (non-self-adjoint coupling through an r -dependent boundary functional), not a gap in effort. What *is* closed form: the per-mode Green’s function, the exact eigen-conditions, and the secular determinant whose numerical roots furnish the spectrum.